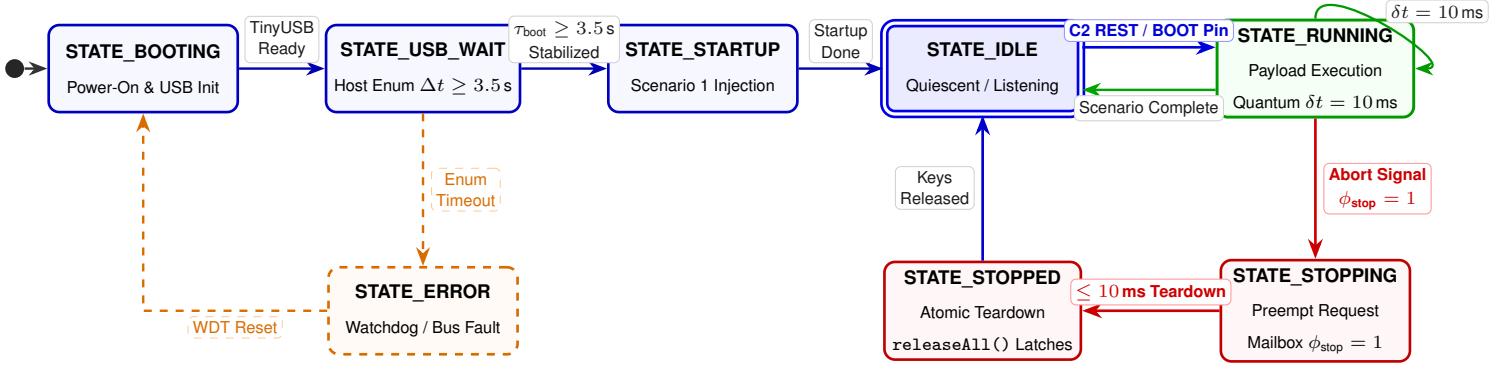

STATE_BOOTING
Power-On & USB Init
TinyUSB Ready
STATE_USB_WAIT
Host Enum $\Delta t \geq 3.5\,\mathrm{s}$
$\tau_{\mathrm{boot}} \geq 3.5\,\mathrm{s}$ Stabilized
STATE_STARTUP
Scenario 1 Injection
Startup Done
STATE_IDLE
Quiescent / Listening
C2 REST / BOOT Pin
STATE_RUNNING
Payload Execution
Quantum $\delta t = 10\,\mathrm{ms}$
$\delta t = 10\,\mathrm{ms}$
Scenario Complete
Enum Timeout
STATE_ERROR
Watchdog / Bus Fault
WDT Reset
Keys Released
Abort Signal
$\phi_{\mathrm{stop}} = 1$
STATE_STOPPED
Atomic Teardown
releaseAll() Latches
$\leq 10\,\mathrm{ms}$ Teardown
STATE_STOPPING
Preempt Request
Mailbox $\phi_{\mathrm{stop}} = 1$